GRADE 7 ACCELERATED · UNIT 6 · LESSON 6

Real-World Coordinate-Plane Problems; Perimeter

Independent Set 6.06 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. Find the perimeter of the triangle $(-8, -3)$, $(2, -3)$, $(-3, 9)$. It has no right angle.

36 units

Base 10; the height 12 lands at (−3, −3) and splits it 5 and 5, so each slanted side is 13.

2. Find the perimeter of the triangle $(-4, 1)$, $(2, 1)$, $(2, -4)$, to the nearest tenth.

18.8 units

6 + 5 + √61 ≈ 18.81.

3. Find the perimeter of the parallelogram $(-4, -5)$, $(2, -5)$, $(7, 7)$, $(1, 7)$.

38 units

Two sides of 6; two slanted sides, across 5 and up 12, of 13.

4. On a map in blocks, Kai's home is at $(-6, -5)$ and his school at $(3, 7)$. How much shorter is the straight line than the streets?

6 blocks

Streets 9 + 12 = 21; straight 15.

5. On a map in miles, a dock is at $(-4, 3)$ and an island at $(3, -5)$. How far apart are they, to the nearest tenth?

√113 ≈ 10.6 mi

Across 7, down 8; 49 + 64 = 113.

6. A rectangle has corners $(-3, -2)$, $(2, -2)$, $(2, 10)$ and $(-3, 10)$. How long is a diagonal?

13 units

Sides 2 − (−3) = 5 and 10 − (−2) = 12; 25 + 144 = 169.